Inflation

- Inflation means that the purchasing power of your money decreases over time. For example, if you had EGP 100k from 10 years, you could buy a house. After 2 years, the same amount of money can only buy a car. After another 2 years, the same amount of money can only buy a motorcycle, and so on. Why does that happen? Because when the economy of a country does bad, the country prints money. When new bank notes are printed, the total number of the bank notes of this currency in the world increases, so their value decreases, so what an EGP 100k could buy from a few years ago can't buy now. This is why we said in the introduction that ***cash loses over time due to inflation***. This is why you should never ever leave cash without investing it.
- Inflation rates differ from one country to another. For example, in Canada and US, the inflation rate is about 2%.
- Inflation is measured using the Consumer Price Index (CPI), which tracks the prices of a basket of goods and services each year (e.g. clothes, transportation, food, etc).
- Your annual salary should increase by at least the inflation rate of the currency you are getting paid in to maintain your purchasing power. If your annual increase is less than the inflation rate, then you are getting poorer over time!

Compounding

- Compounding your interest is the process of re-investing the interest earned on your investment so that during the next period, you earn interest not only on your invested capital, but also on the accumulated interest.
- Let's take an example to see the power of compounding. Assume you have $10,000 that you want to invest for 20 years, and you have an investment opportunity that gets you 10% each year. You have two choices:
    1. No Compounding: each year, you take the 10% profit and spend them on whatever you want.
    2. Compounding: each year, you keep this 10% profit for investment.
- Check the shocking analysis below. On the left is the no-compounding case. At the end of the 20 years, you would have $30,000. On the right is the compounding case. At the end of the 20 years, you would have $67,275! See the difference? Notice that in both cases you are leaving the $10,000 for 20 years. So, what made this huge difference? You just kept the profit every year for more investments instead of taking it! So, in the no-compounding case, you get $1000 for 20 years. Then, after the 20th year, you get $10,000, which is your original invested capital. In the compounding case, you get nothing for 20 years. Then, after the 20th year, you get $67,275.

	initial deposit	interest			initial deposit	interest	new deposit
0	$ 10,000.00			0	$ 10,000.00		
1		$1,000.00		1		$1,000.00	$11,000.00
2		$1,000.00		2		$1,100.00	$12,100.00
3		$1,000.00		3		$1,210.00	$13,310.00
4		$1,000.00		4		$1,331.00	$14,641.00
5		$1,000.00		5		$1,464.10	$16,105.10
6		$1,000.00		6		$1,610.51	$17,715.61
7		$1,000.00		7		$1,771.56	$19,487.17
8		$1,000.00		8		$1,948.72	$21,435.89
9		$1,000.00		9		$2,143.59	$23,579.48
10		$1,000.00		10		$2,357.95	$25,937.42
11		$1,000.00		11		$2,593.74	$28,531.17
12		$1,000.00		12		$2,853.12	$31,384.28
13		$1,000.00		13		$3,138.43	$34,522.71
14		$1,000.00		14		$3,452.27	$37,974.98
15		$1,000.00		15		$3,797.50	$41,772.48
16		$1,000.00		16		$4,177.25	$45,949.73
17		$1,000.00		17		$4,594.97	$50,544.70
18		$1,000.00		18		$5,054.47	$55,599.17
19		$1,000.00		19		$5,559.92	$61,159.09
20		$1,000.00		20		$6,115.91	$67,275.00
21	$ 30,000.00			21	$ 67,275.00		